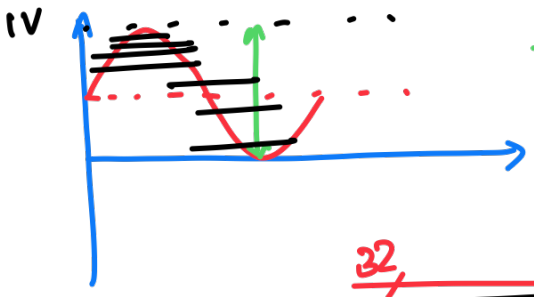
Lecture #37.

FLASH ADC.

Ex: 10-bit ADC

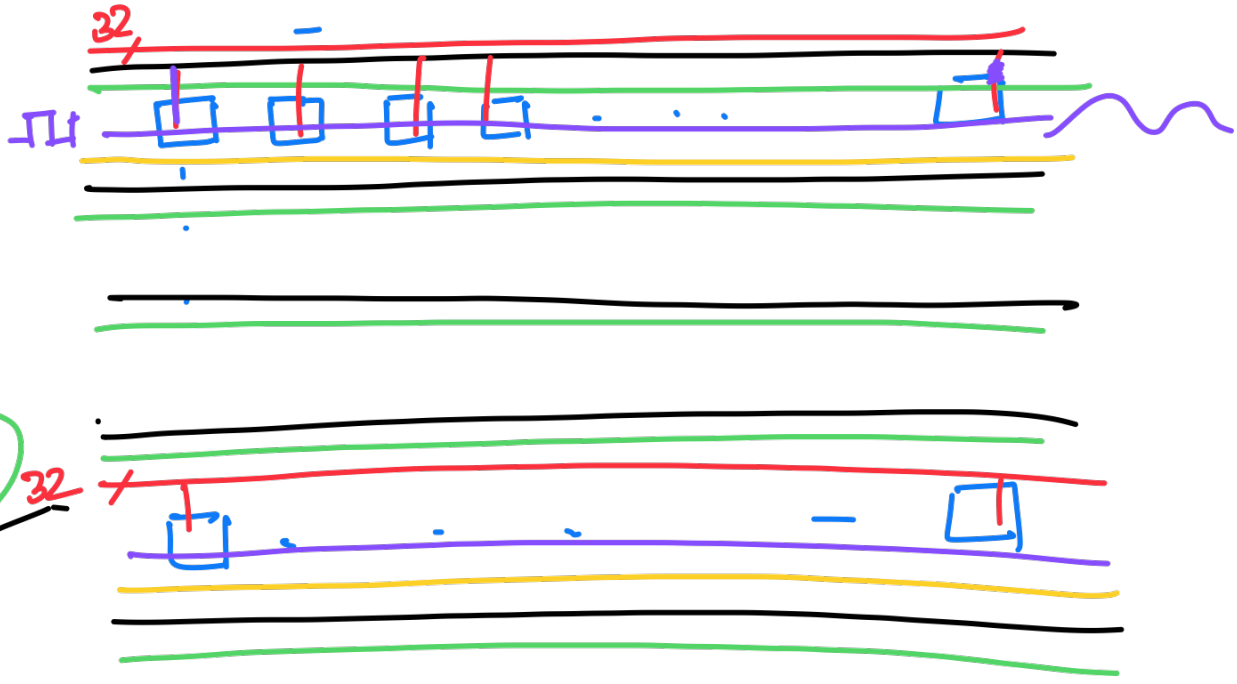


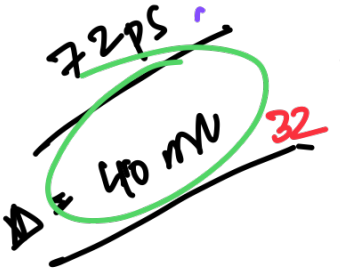
- $\Delta = \frac{1}{2^{10}} = \frac{1}{1024} V \Rightarrow V_{th}$ (threshold voltage)

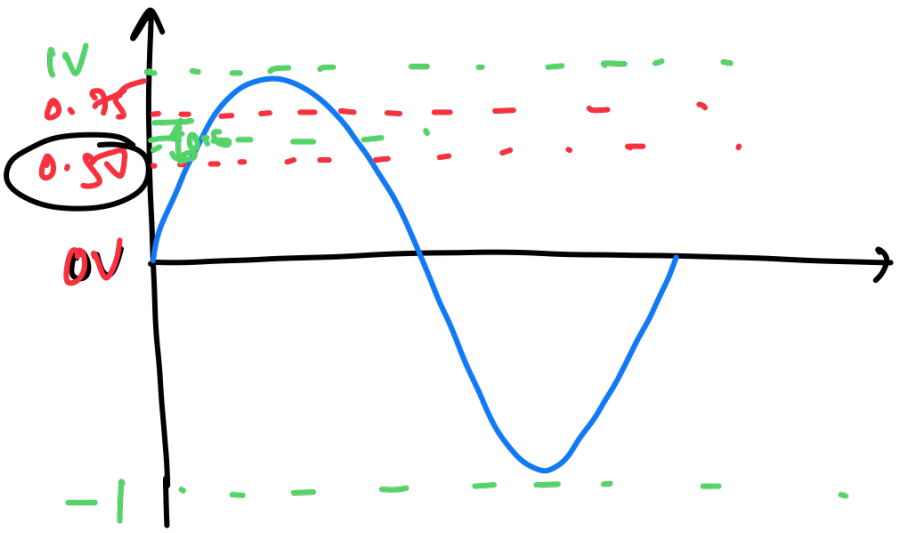
- # of comparators = $2^{10} - 1 = 1023$

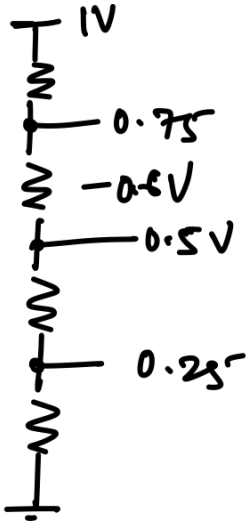
14 GS/s.

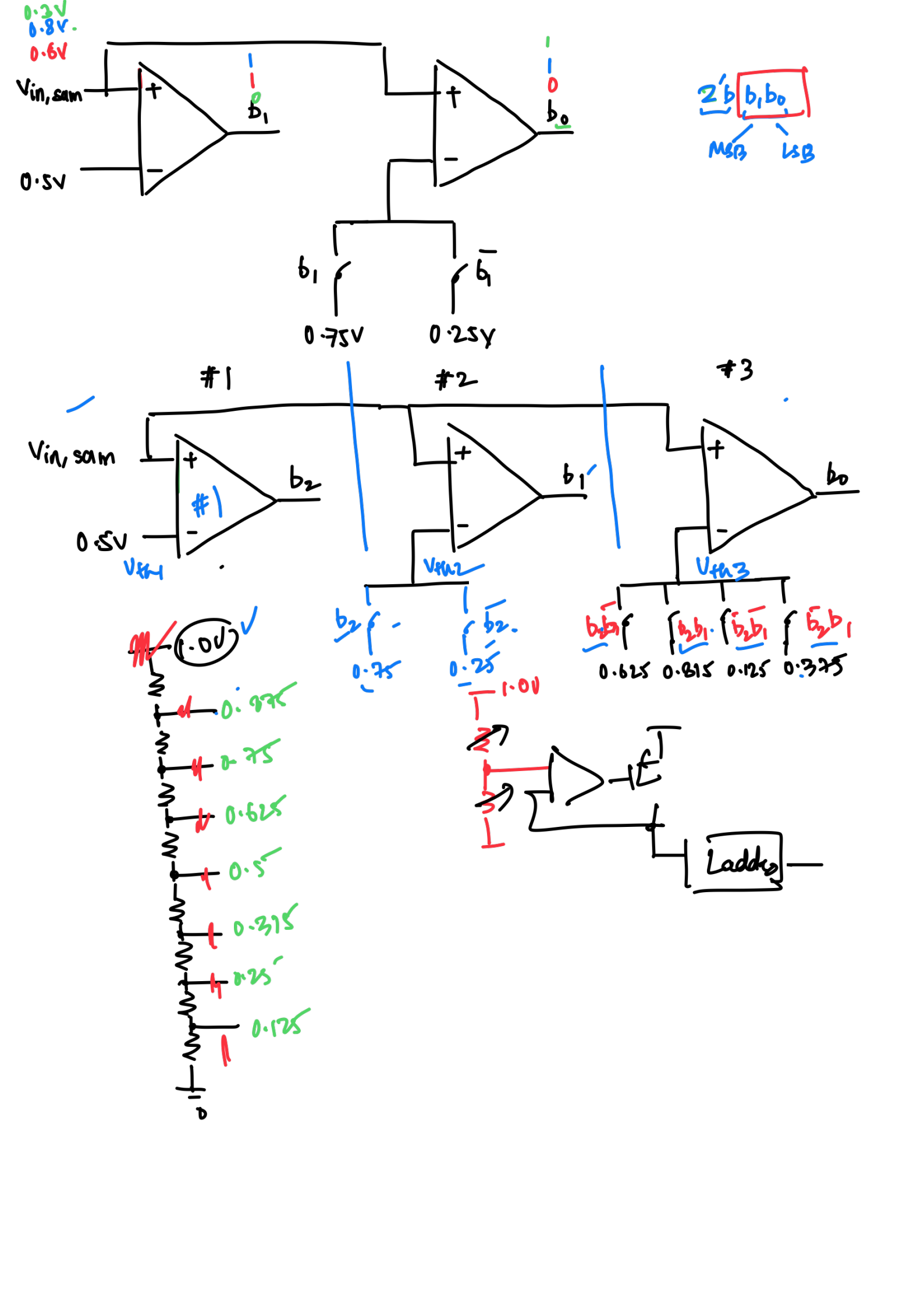
4-bit ADC



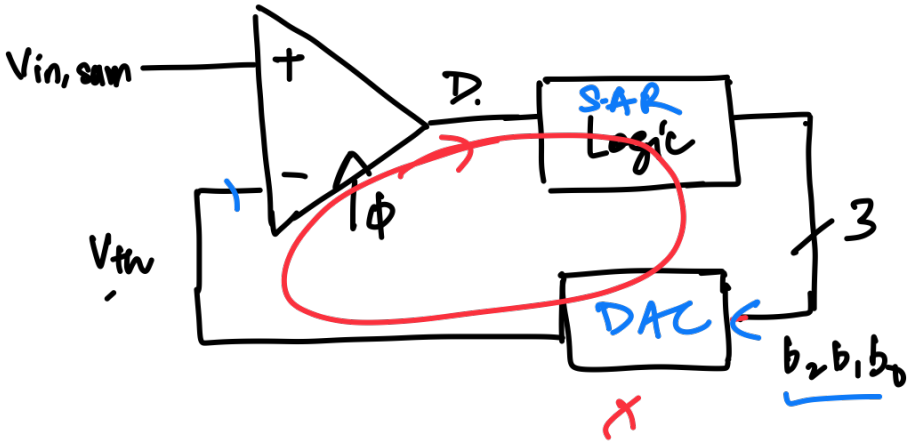






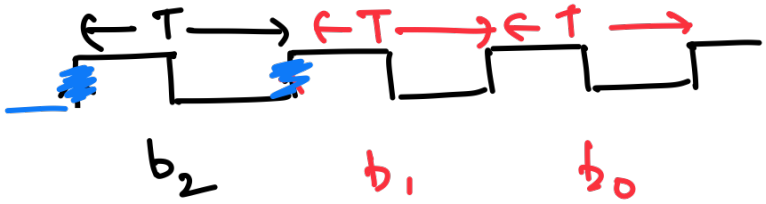

0.3V
0.8V
0.6V
Vin, sam
0.5V
b1
b0
2'b b1b0
MSB
LSB
b1
b1'
0.75V
0.25V
#1
#2
#3
Vin, sam
0.5V
b2
b1'
b0
Vref1
Vref2
Vref3
b2
b2'
0.75
0.25
b2b1
b2b1'
b2'b1
b2'b1'
0.625
0.875
0.125
0.375
1.0V
1.0V
0.875
0.75
0.625
0.5
0.375
0.25
0.125
0
Ladder

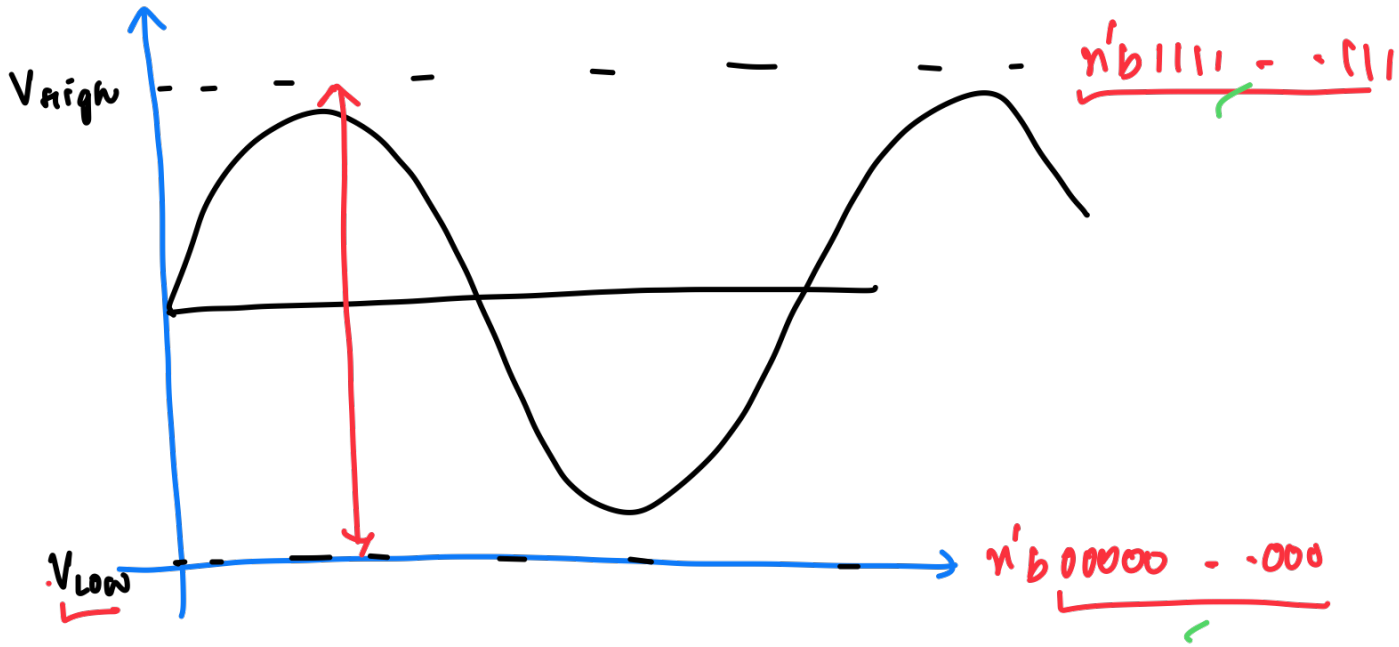
— SAR ADC

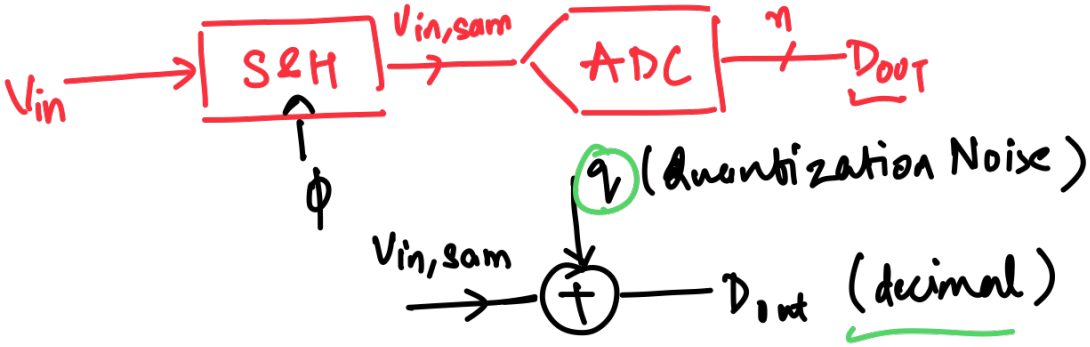


DAC: Digital to Analog Converter

b_2	b_1	b_0	
0	0	0	t<0
1/0	0	0	T
1/0	1/0		2T







— $0 \le q < \Delta$ — $-\Delta \le q < 0$

— $-\frac{\Delta}{2} \le q < \frac{\Delta}{2}$

- e_q as random variable
- q has uniform distribution.

$$q \in \left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right) \quad \text{with} \quad pdf(q) = \frac{1}{\Delta}$$

$$E[q] = 0$$

$$E[q^2] = \frac{\Delta^2}{12} = \int_{-\frac{\Delta}{2}}^{\Delta/2} q^2 \cdot f(q)\, dq$$

- $V_{in} = A\sin(\omega_0 t)$

$$\overline{V_{in}^2} = \frac{A^2}{2}$$

- Signal to Noise ration $(SNR) = 10\log_{10}\left(\frac{\text{Sig. Pow}}{\text{Noise Pow}}\right)$

$$SNR = 10\log_{10}\left(\frac{A^2/2}{\Delta^2/12}\right)$$